

Title

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05/30/2026



OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
 - Visualization – Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion
- Appendix



EXECUTIVE SUMMARY



- Trends in Programming Languages
 - JavaScript remains the most commonly used
 - Python is the fastest-growing
- Demographics
 - Predominantly young
 - Talent is heavily concentrated in the United States, India, and Germany
- Compensation
 - Average salary is around \$100,000
 - Swift demands higher than average pay
- Location
 - Washington, DC is a top-tier city for job posting volume



INTRODUCTION



This report will analyze the current talent landscape and technical shifts within the developer community to identify future skill requirements.

- Adoption of emerging tools to stay competitive
- Evaluation of global demographics and compensation benchmarks

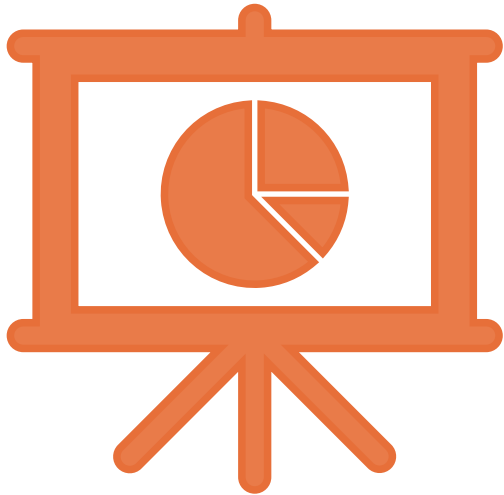
METHODOLOGY



- Data Source: The primary data source for this analysis is the Survey dataset. This comprehensive survey captured data on developers covering everything from specific programming languages used to demographic information and salary details.
- Data Collection
 - Survey Responses
 - Technical Metrics
 - Demographic & Economic Data
- Data Wrangling
 - Handling Missing Values
 - Standardization
 - Data Transformation
 - Filtering & Aggregation
 - Categorization



Results



<Please present your results in the following slides.>



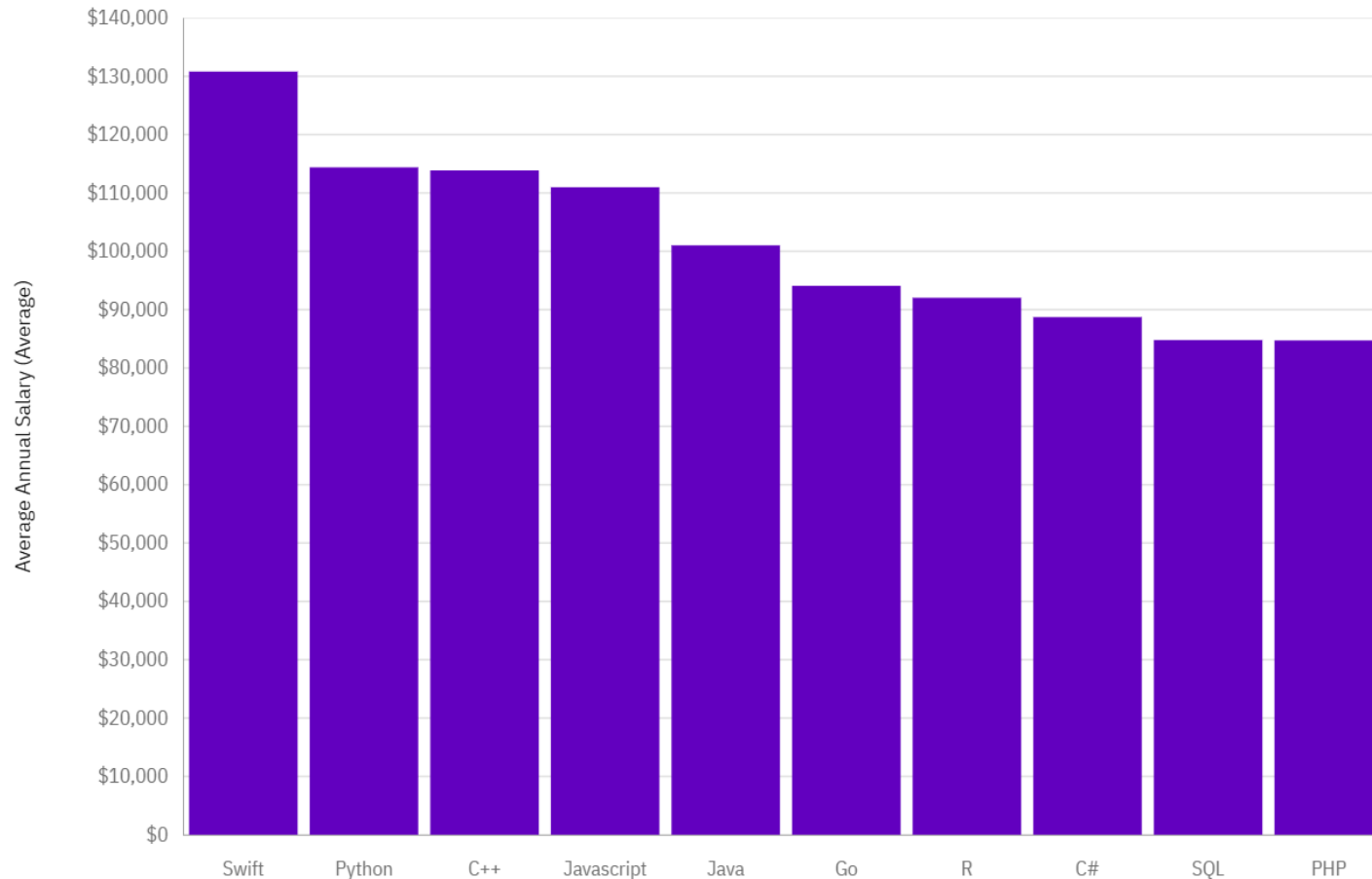
JOB POSTINGS

- In Module 1 you have collected the job posting data using Job API in a file named “job-postings.xlsx”. Present that data using a bar chart here. Order the bar chart in the descending order of the number of job postings.



POPULAR LANGUAGES

Average Annual Salary by Language



In Module 1 you have collected the job data into a file named “popular-languages.csv”. Prepare a bar chart in the descending order of salary.

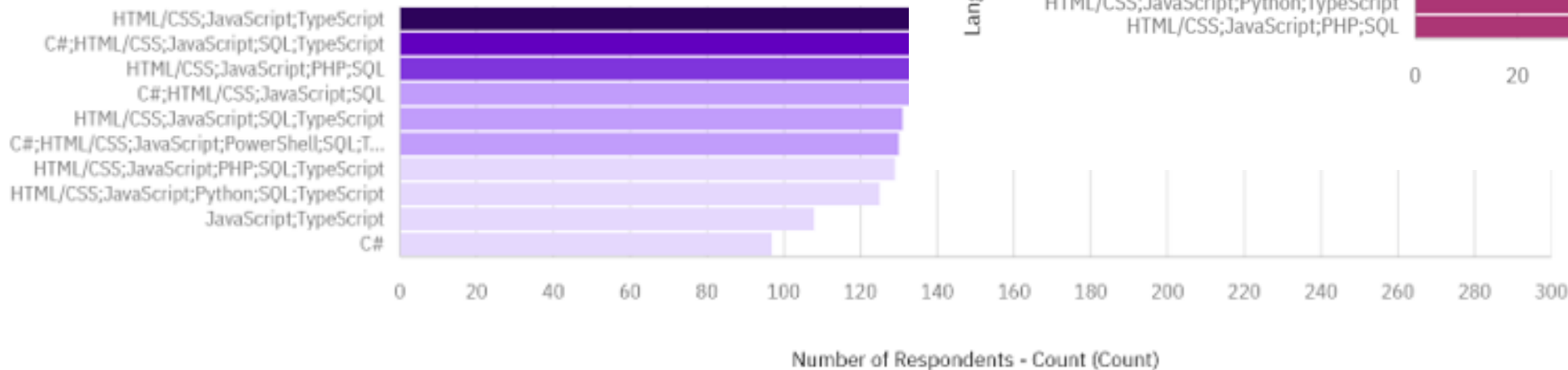
PROGRAMMING LANGUAGE TRENDS

Summarize key trends shown in the charts

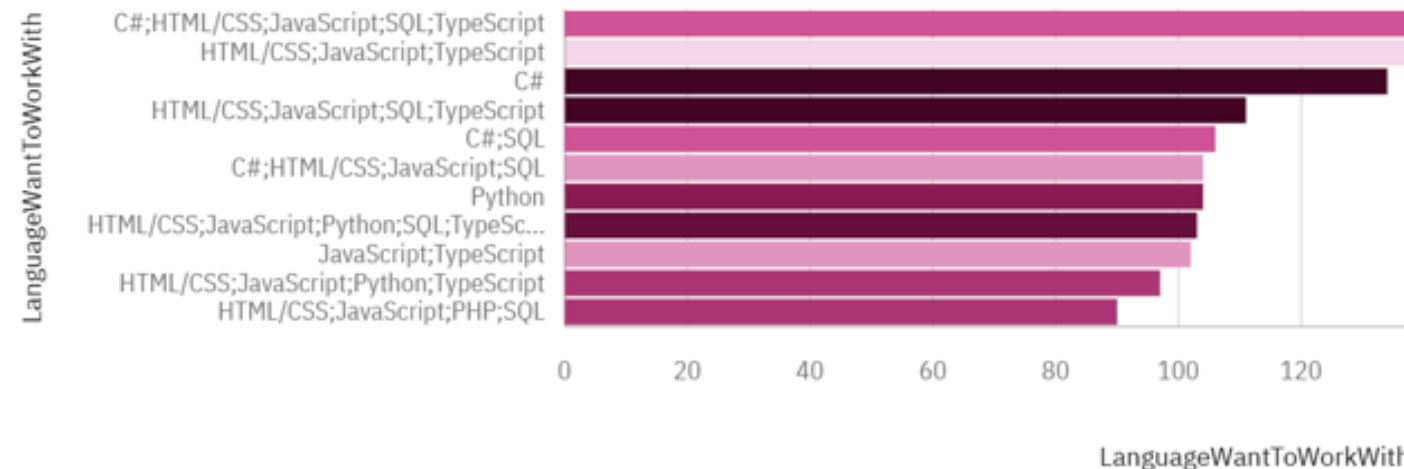
Current Year

<Bar chart of top 10 programming

Top 10 Programming Languages Have W



Top 10 Languages Want to Work With



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Java remains at the top
- C+ Increases next year
- Python increases next year

Implications

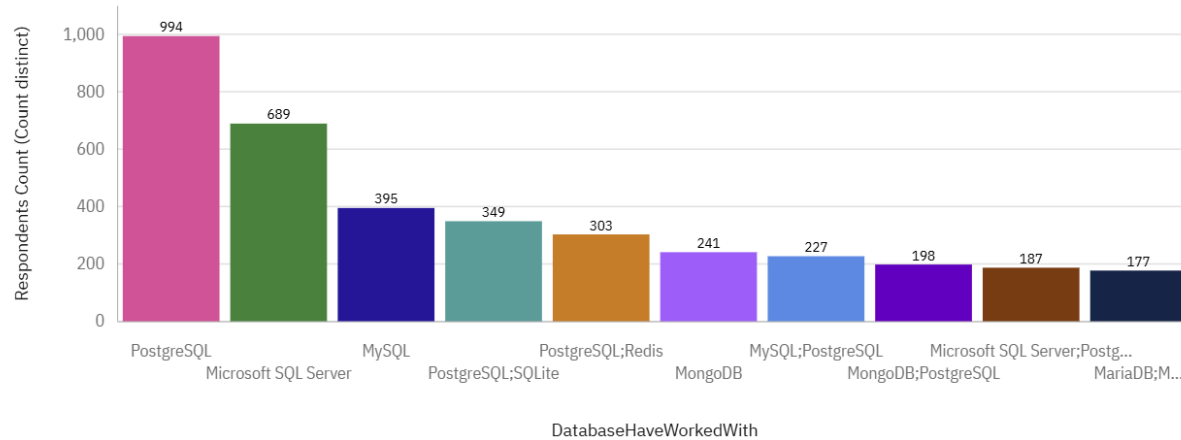
- Web developers are still in high demand
- Java will remain relevant
- Rise in AI will affect recruitment

DATABASE TRENDS

Summarize key trends shown in the charts.

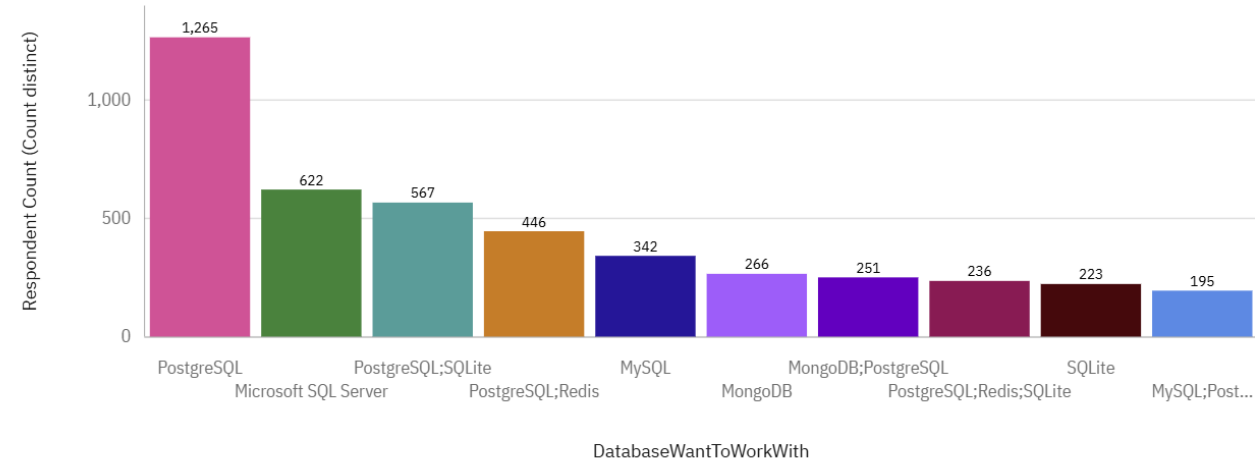
Current Year

Top 10 Databases Have Worked With



Next Year

Top 10 Databases Want to Work With



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL leads in both the current year and next year
- MySQL decreases next year
- MongoDB increases next year

Implications

- Investment in PostgreSQL carries a low risk
- Open-source is still the overall preference
- No overall major swings

DASHBOARD

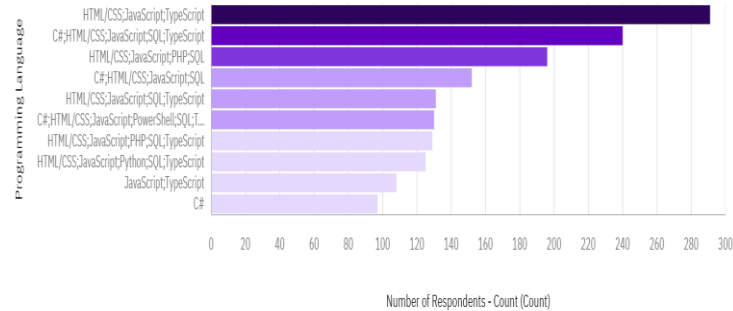


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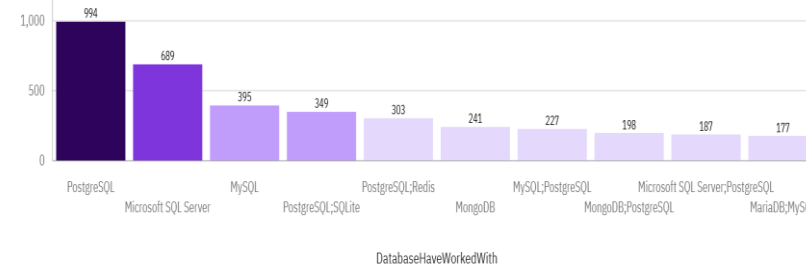


DASHBOARD TAB 1: Current Technology Usage

Top 10 Programming Languages Have Worked With



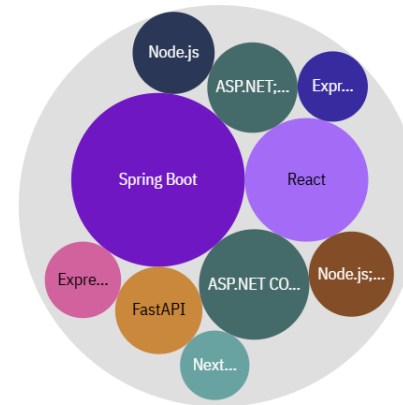
Top 10 Databases Have Worked With



Top 10 Platforms Worked With



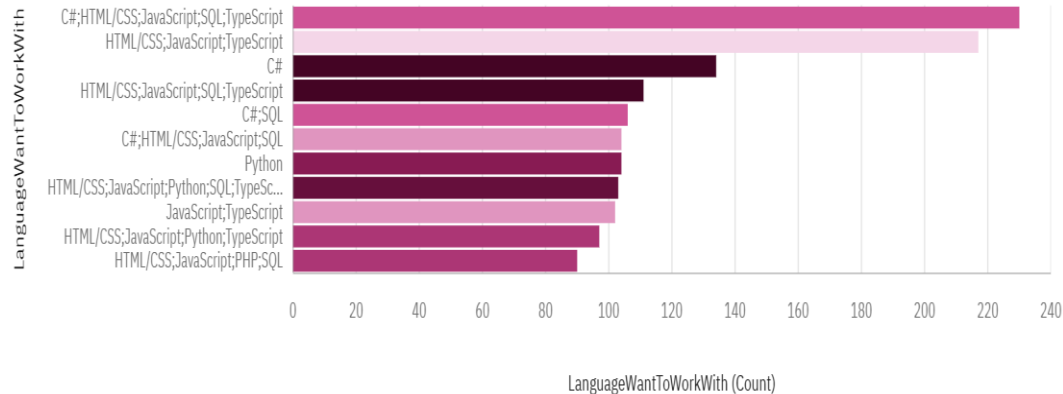
Top 10 Web Frames Worked With



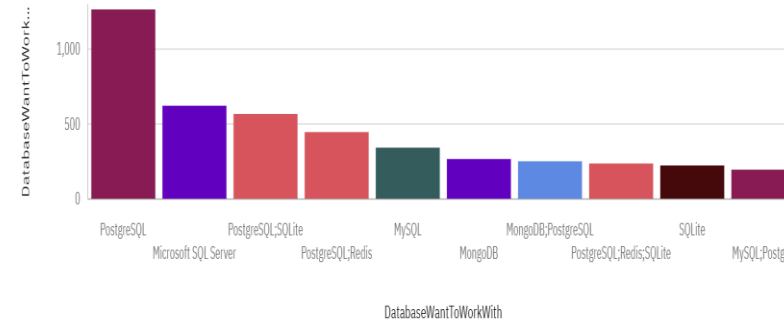
- JavaScript is the top language
- PostgreSQL is the top database
- AWS is the top platform
- SpringBoot is the top web framework

DASHBOARD TAB 2: Future Technology Trends

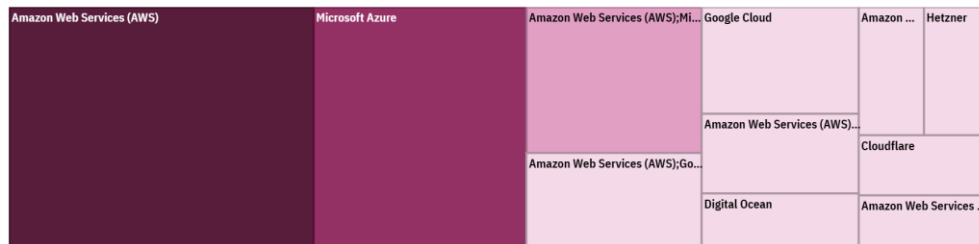
Top 10 Languages Want to Work With



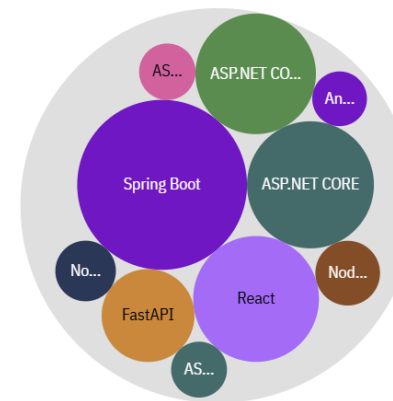
Top 10 Databases Want to Work With



Top 10 Platforms Want to Work With



Top 10 Webframes Want to Work With

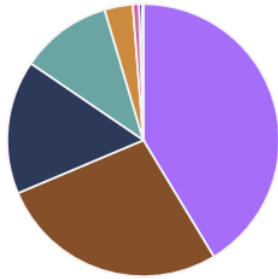
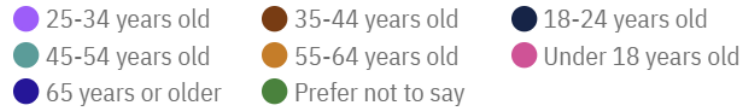


- JavaScript remains the top language
- PostgreSQL remains the top database
- AWS remains the top platform
- SpringBoot remains the top webframe

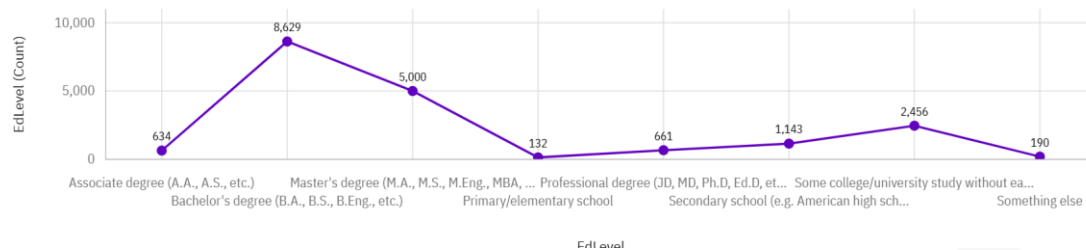
DASHBOARD TAB 3: Demographics

Responses by Age

Age

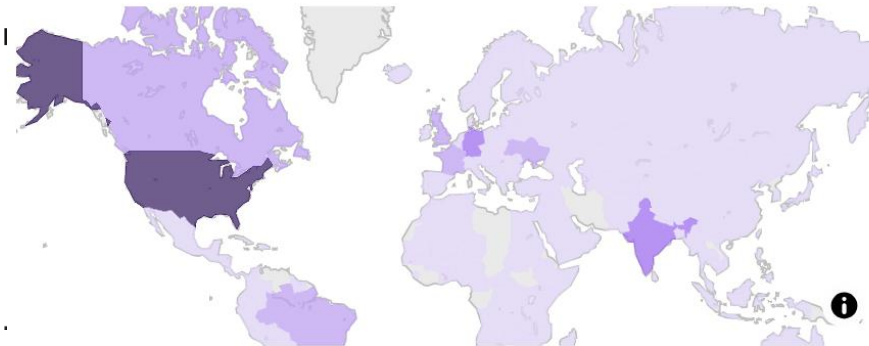


Responses by Education Level



Respondents by Country

ResponseId (Count)

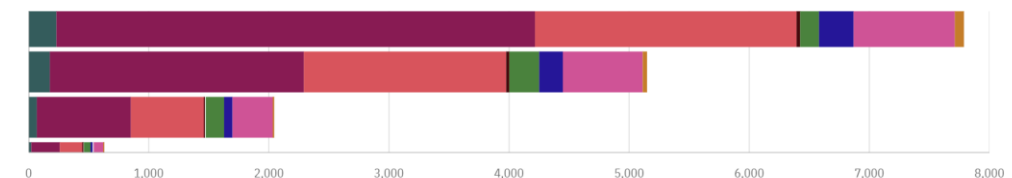


Respondent Count by Age classified by Educational Level

EdLevel

Legend for Respondent Count by Age classified by Educational Level:

- Associate degree (A.A., A.S., etc.) (Dark Green)
- Bachelor's degree (B.A., B.S., B.E... (Dark Red)
- Master's degree (M.A., M.S., M.En... (Red)
- Professional degree (JD, MD, Ph.D... (Green)
- Secondary school (e.g. American ... (Blue)
- Some college/university study wit... (Pink)
- Primary/elementary school (Brown)
- Something else (Orange)



- Most respondents are between 25-34 years old
- Most respondents are from the US



DISCUSSION



- The majority of the workforce has a Bachelor's degree and are between the ages of 25 and 34 years old
- PostgreSQL is the top database in both the current year and next year
- JavaScript is the top language in both the current year and next year
- No significant changes next year

OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript is the top in both current and next year for languages
- PostgreSQL is the top in both current and next year for databases
- AWS is the current leader in platforms

Implications

- Prioritization goes with JavaScript
- Database prioritization goes to PostgreSQL
- More companies in the platform space will put pressure on AWS

CONCLUSION



- Trends for current and next year stay consistent
- Need to look beyond next year to stay competitive
- Workforce is young and educated



APPENDIX



- IBM Cognos Analytics
- Jupyter Notebooks
- Stack Overflow Developer Survey

